DIBBs eCR Viewer

User Acceptance Testing Guide – End Users

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Version History

Version	Author	Reviewed By	Revision Date	Revision Notes
1.0	Elizabeth Chinelo Ikejimba	Kristie Clarke	09/17/2025	Revised to contain details for both NBS-integrated Viewer and Standalone eCR Library
1.1	Matt Goldberg	Kristie Clarke	12/10/2025	Revised based on Kristie's feedback
1.2	Matt Goldberg	Michael Wodajo	04/23/2026	Revised to update to current information

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1 Objectives

Note: For definitions of any unfamiliar terms or acronyms (e.g., eCR), refer to the Glossary at the end of this document.

This guide lays out the steps for case investigators and other end users of eCR documents to conduct user acceptance testing (UAT) for the eCR Viewer.

In this guide, we refer to two versions of the product:

1. Integrated eCR Viewer
2. Standalone eCR Library

Following this guidance helps ensure either version of the eCR Viewer is behaving as expected and enables users to identify any bugs that need to be addressed.

2 Background

The eCR Viewer is a free, open-source application offered by CDC that make eCR data more usable and useful for case investigators, epidemiologists, and others working with eCR data.

3 Testing Scope

The Integrated eCR Viewer allows jurisdictions to view all received eCRs in their surveillance system (e.g., NBS and EpiTrax) and open individual records for detailed review. The Standalone eCR Library allows jurisdictions to view all received eCRs in a dashboard and open individual records for detailed review. For UAT, we want to answer two main questions:

1. Do all features work as expected?
2. Do the eCR Viewer and the eCR Library fully convert all data from the source eCRs? In other words, is the data quality high?

4 Test Plan

4.1 Prerequisites

To begin UAT, your eCR coordinator will share:

- ☐ Access to the eCR Viewer and the eCR Library (if applicable)
- ☐ Several test eCR & RR files (HTML documents)
- ☐ The End Users UAT Checklist

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If you're missing any of these resources, please contact your eCR coordinator.

4.2 Testing Flow

Use the End Users UAT Checklist to confirm completion of specific tasks for each of the steps below. Make sure you pull it up for reference.

You'll complete testing in three steps:

4. Test the full feature set of the eCR Viewer and the eCR Library (if applicable).
5. Check the feature functionality of the individual eCR document view.
6. Assess the data quality of each assigned test eCR.

4.3 Library Feature Testing

If you only have the NBS-integrated eCR Viewer, then skip to section "4.4 Individual Record Feature Testing".

To test the feature functionality of the eCR Library, open the End Users UAT Checklist.

Navigate from the "Full Feature Set" tab to open the "Feature Functionality" tab.

1. Open the eCR Library.
2. Verify the following by marking Yes or No:
    a. Filtering by condition only shows eCRs with that reportable condition present
    b. Filtering by date only shows eCRs within the specified date range
    c. Searching by patient name only shows patients who match the search query
    d. Pagination (moving from page 1-2, changing number of eCRs displayed, etc.) works as expected
    e. Sorting by name and dates works as expected
    f. Where applicable, eCRs are grouped by version
    g. Program management works as expected
    h. User management works as expected

4.4 Individual Record Feature Testing

Next, we'll check feature functionality when viewing an individual eCR document. You should still be on the "Feature Functionality" tab of your checklist.

1. Open an eCR — any eCR will work here.
    a. From the eCR Library (if you have access to it), click the patient name from the table.
    b. From NBS/EpiTrax (if you have access to it), click the View eCR button.
2. Verify the following by marking Yes or No:
    a. The patient banner at the top of the page displays the patient's name and date of birth, and is sticky as you scroll the page
    b. Side navigation is functional — clicking a tab navigates you to that section of the page
    c. Accordions on the page expand and collapse as expected
    d. The Unavailable Information section is present at the bottom of the document

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4.5 Data Quality Testing

Finally, we'll check data quality of each eCR by comparing each field in the original eICR/RR to the data display in the eCR Viewer and eCR Library and identifying any disparities.

4.5.1 NBS-integrated eCR Viewer

1. Open the "Data Quality" tab of the End Users UAT Checklist for this test. Open an original (HTML) eCR and note the patient name and eCR document ID number.
2. Find that same patient and document in NBS and open the eCR document.
3. To verify the document IDs match, navigate to eCR Metadata > eICR Details > eICR ID within the Viewer and compare it against the document ID from the original eCR.
4. Using the original eCR as your source of truth, check to ensure every data element is also present in the Viewer's eCR document. Consult the End Users UAT Checklist for specific fields. You'll want to check if:
    a. Any fields are missing entirely from the eCR document
    b. Any fields are missing a portion of data from the eCR document (e.g., emergency contact name is present but is missing phone number)
    c. Any fields may be formatted unexpectedly or incorrectly (especially tables)
5. Repeat for every eCR your coordinator has shared.

4.5.2 Standalone eCR Library

1. Open the "Data Quality" tab of the End Users UAT Checklist for this test. Open an original (HTML) eCR and note the patient name and eCR document ID number.
2. Find that same patient and document in the eCR Library. Check that all expected fields are present and open the eCR document.
3. To verify the document IDs match, navigate to eCR Metadata > eICR Details > eICR ID within the Viewer and compare it against the document ID from the original eCR.
4. Using the original eCR as your source of truth, check to ensure every data element is also present in the eCR document within the eCR Library. Consult the checklist for specific fields. You'll want to check if:
    a. Any fields are missing entirely from the eCR document
    b. Any fields are missing a portion of data from the eCR document (e.g., emergency contact name is present but is missing phone number)
    c. Any fields may be formatted unexpectedly or incorrectly (especially tables)
5. Repeat for every eCR your coordinator has shared.

4.6 More Information about Data Quality

The original eCR documents (i.e., the HTML files) are views of the raw eICR, formatted as XML C-CDA. The eCR Viewer and the eCR Library take this same raw file and converts it to a different standard for internal use.

As a result, some data that appears in the Viewer and the Library isn't in the original HTML, and some data visible in the HTML isn't shown in the eCR document found within the Viewer and the Library. Once the missing data is identified, the DIBBs team will work to add coverage for those fields.

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4.7 Wrapping Up

Finally, you'll share the completed End Users UAT Checklist back to your eCR coordinator. They will share this feedback with the DIBBs team for further action and issue resolution.

Thank you so much for participating in the UAT process! Your feedback will help ensure that the eCR Viewer and the eCR Library are as robust as they can possibly be.

If you have any questions or want to chat with us directly, please reach out at dibbs@cdc.gov.

5 Glossary

API: Application programming interface (API) is a set of functions and procedures that allow the creation of applications that access features and or data of another operating system, application, or other service.

eCR: electronic case reporting (eCR) is the automated, real-time exchange of case report information between electronic health records (EHRs) and public health agencies.

eICR: electronic initial case report (eICR) sent from a clinical care provider to determine if the condition code is reportable for its jurisdiction.

EHR: electronic health records (EHR) are the electronic version of a patient's medical history that providers maintain over time and may include all of the key administrative clinical data relevant to that patient's care.

eRSD: electronic reporting and surveillance distribution service (eRSD) is a service provided by APHL that looks at the EHR to determine if there's a trigger code present by using LOINC codes.

LOINC: Logical observation identifiers names and codes (LOINC) is a standardized coding system used for identifying health measurements, observations, and services. LOINC codes facilitate the exchange and aggregation of clinical laboratory data.

NBS: NEDSS Base System (NBS) is a CDC-developed integrated information system that helps STLT public health departments manage reportable disease data and send notifiable disease data to the CDC.

NEDSS: National Electronic Disease Surveillance System (NEDSS) is a set of architectural standards to support NNDSS.

RCKMS: Reportable Conditions Knowledge Management System (RCKMS) is an authoritative, real-time, digital portable to enhance disease surveillance by providing comprehensive information on public health reporting requirements to public health reporters such as clinicians and laboratories.

RR: Reportability response (RR) is a document, sent from a jurisdiction to a clinical care provider, that communicates information on the status of a reportable condition/

Schema: A schema is how a database categorizes recorded data.

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STLT: State, tribal, local, and territory (STLT)

SQL: Structured query language (SQL) is a programming language for interacting with a database.

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